PROGRAM 5: Speech-to-Text for Engineering Queries

AIM

To convert spoken engineering-related queries into text using a pretrained speech recognition model.

ALGORITHM

1. Import the required libraries.
2. Load a pretrained Whisper speech recognition model.
3. Upload an audio file.
4. Process the audio using the model.
5. Display the converted text.

CODE

Use an English audio recording in WAV, MP3 or another supported format.

```
from transformers import pipeline
from google.colab import files

recognizer = pipeline(
    "automatic-speech-recognition",
    model="openai/whisper-tiny"
)

uploaded = files.upload()

audio_file = next(iter(uploaded))

result = recognizer(audio_file)

print("\nRecognized Text:")
print(result["text"])
```

INPUT

Upload an audio recording containing:

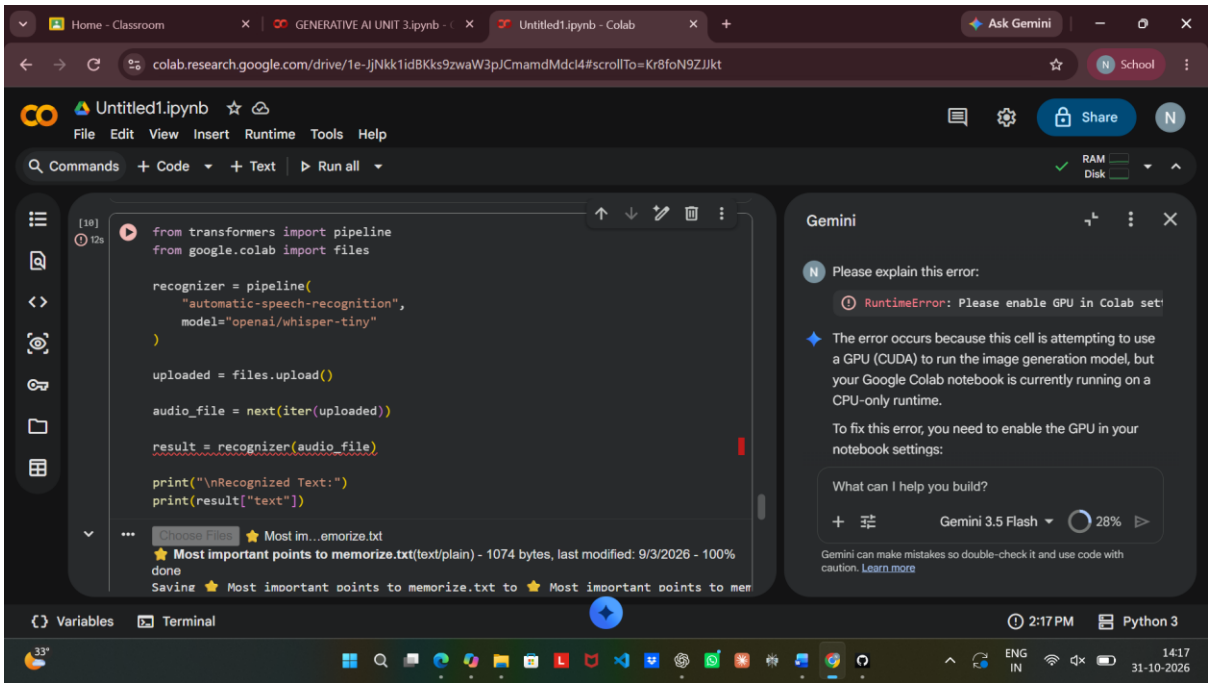
What is artificial intelligence?

OUTPUT

Recognized Text:

What is artificial intelligence?

The recognized text may differ slightly depending on pronunciation and audio quality.



RESULT

Thus, spoken engineering queries were converted into text using a pretrained Whisper model.